

CSE 370– Database Systems

QUIZ 3 (Set B)

Spring' 25

Marks

Full Name (in Block Letter): _____

ID: _____ **Section:** ____ **Signature:** _____

Date: _____ **Total Marks: 10** **Time: 10 minutes**

Consider the following relation:

Book_Lending (libraryID, bookID, copyNo, loanID, libraryName, branchLocation, establishedYear, lateFee, genre, pageCount, bookTitle, authorID, authorName, authorEmail, borrowerRating, daysBorrowed)

Primary Key: (libraryID, bookID, copyNo, loanID)

Functional Dependencies:

- **FD1:** branchLocation, establishedYear → lateFee
- **FD2:** authorID → authorName, authorEmail
- **FD3:** libraryID, bookID, copyNo, loanID → libraryName, branchLocation, establishedYear, lateFee, genre, pageCount, bookTitle, authorID, authorName, authorEmail, borrowerRating, daysBorrowed
- **FD4:** bookID → genre, pageCount, bookTitle
- **FD5:** libraryID → libraryName, branchLocation, establishedYear, lateFee
- **FD6:** bookID, copyNo, loanID → authorID, authorName, authorEmail

(a) Explain if the above relation is in the **first normal form (1NF)** or not. If not, apply **1NF normalization**.

(b) Explain if the relation(s) from part (a) is/are in the **second normal form (2NF)** or not. If not, apply **2NF normalization**.

(c) Explain if the relation(s) from part (b) is/are in the **third normal form (3NF)** or not. If not, apply **3NF normalization**.